

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Computer Vision with OpenCV

COURSE CODE: DSA0202

COURSE OUTCOME COVERED

CO4: Evaluate recognition and learning methods including machine learning and deep learning models (CNNs, GANs, SVM, HMM, etc.) for vision tasks.. (BL5)

Assessment Tool Weightage: 30%

QUESTIONS: 2

Total Marks:30

Annexure A

Industry Problem Solving Task

Code of Conduct:

I P Sai Bhavyatha(192424333)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Palacharla Sai Bhavyatha

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context	
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts	
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution	
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools	
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis	
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation	

Industry Problem Solving Task

1: Automated Defect Detection in Manufacturing

A precision engineering company wants to automatically detect defects in mechanical components using computer vision. Parts vary slightly in shape due to manufacturing tolerances.

Questions:

1. How can PCA-based shape priors and contour-based representations be used to model standard component shapes?
2. Propose a pipeline using machine learning (SVM, GMM) or deep learning (CNN) to classify defective vs. non-defective parts.
3. Discuss methods to handle noise, partial occlusions, and varying illumination in industrial images.

2: Autonomous Vehicle Obstacle Recognition

A self-driving car company needs to detect rare or unexpected obstacles on urban roads. The labeled data for these obstacles is limited.

Questions:

1. Explain how data augmentation using GANs can help generate synthetic images for training.
2. Compare the effectiveness of CNNs versus Vision Transformers for detecting small or sparse objects in high-resolution images.
3. Suggest a real-time inference pipeline for obstacle detection, considering computational constraints in embedded systems.

ANSWERS

1. Automated Defect Detection in Manufacturing

Industry Problem

A precision engineering company manufactures mechanical components where small defects such as cracks, scratches, dents, holes, deformation, and surface irregularities must be detected automatically. Since the components may vary slightly because of manufacturing tolerances, a rigid template-matching approach may incorrectly classify acceptable parts as defective. Therefore, a computer vision system should learn the normal variation of component shapes and distinguish genuine defects from permissible manufacturing variations.

Question 1: PCA-Based Shape Priors and Contour-Based Representations

Answer

PCA-based shape priors can be used to learn the normal shape variation of mechanical components from a collection of defect-free training images.

First, images of correctly manufactured components are collected under controlled conditions. The component boundary is extracted using edge detection or contour extraction techniques. Important contour points or landmarks are selected and represented as numerical coordinates.

The training shapes are aligned to remove differences caused by translation, rotation, and scale. PCA is then applied to the aligned shape vectors.

The shape can be represented as:

$$\mathbf{S} = \bar{\mathbf{S}} + \mathbf{P}\mathbf{b}$$

where:

- $\mathbf{S}$ = reconstructed component shape
- $\bar{\mathbf{S}}$ = mean shape of defect-free components
- $\mathbf{P}$ = principal component matrix
- $\mathbf{b}$ = shape variation parameters

PCA captures the major variations that occur naturally because of manufacturing tolerances. A new component can then be compared with the learned normal shape model.

If the shape variation of a test component falls within the acceptable PCA range, it can be considered a normal component. If its contour deviates significantly from the learned shape model, it may indicate deformation or another structural defect.

Contour-Based Representation

Contour-based representations describe the outer boundary of the component. Techniques such as **Canny edge detection, contour extraction, Fourier descriptors, curvature analysis, and shape descriptors** can be used.

Important features include:

- Contour length
- Area
- Perimeter
- Aspect ratio
- Curvature
- Corner locations
- Fourier descriptors
- Distance from the mean contour

Conclusion

PCA provides a statistical model of acceptable shape variations, while contour-based features describe the geometry of the component. Combining both methods allows the system to distinguish manufacturing tolerances from genuine shape defects more reliably.

Question 2: Machine Learning / Deep Learning Pipeline

Answer

A complete automated defect detection pipeline can use **SVM, GMM, or CNN** depending on the available dataset and computational resources.

Proposed Pipeline

Image Acquisition → Preprocessing → Component Segmentation → Feature Extraction → Classification → Defect Localization → Quality Decision

Step 1: Image Acquisition

Industrial cameras capture images of mechanical components. Consistent camera position, focus, and lighting should be maintained.

Step 2: Preprocessing

The captured images are preprocessed to remove unwanted noise and normalize illumination.

Operations may include:

- Image resizing
- Grayscale conversion
- Gaussian filtering
- Histogram normalization
- Contrast enhancement
- Background removal

Step 3: Component Segmentation

The component is separated from the background using thresholding, edge detection, or segmentation methods.

Step 4: Feature Extraction

For an **SVM/GMM-based approach**, features such as the following can be extracted:

- Shape descriptors

- Contour features
- Texture
- Edges
- Histogram features
- PCA coefficients
- Local binary patterns

Step 5: Classification

SVM:

An SVM classifier can classify components into **Defective** and **Non-Defective** classes based on extracted features. It is particularly suitable when the dataset is relatively small and features are well defined.

GMM:

A Gaussian Mixture Model can model the probability distribution of normal and defective feature patterns. It can be useful when the feature distributions contain multiple clusters.

CNN:

A CNN can learn image features automatically and directly classify the component as defective or non-defective. For larger labeled datasets, CNN-based classification or object-detection models can provide better performance.

Step 6: Defect Localization

If the exact defect location is required, an object-detection or segmentation model can identify the position and size of the defect.

Step 7: Final Quality Decision

The system produces a final output:

Normal Component → Accept

Defective Component → Reject / Human Inspection

Recommended Approach

For a small industrial dataset, an **SVM + PCA/contour features** approach is practical and computationally efficient.

For a sufficiently large labeled image dataset, a **CNN-based approach** is generally more suitable because it can automatically learn complex visual features.

Conclusion

The proposed pipeline combines image preprocessing, shape analysis, feature extraction, and machine learning/deep learning classification. It can reduce manual inspection effort, provide consistent defect detection, and support real-time quality control.

Question 3: Handling Noise, Partial Occlusions and Varying Illumination

Answer

Industrial images may contain noise caused by cameras, reflections, dust, shadows, surface texture, or changing illumination. Therefore, robust preprocessing is essential.

1. Handling Noise

Noise can be reduced using:

- Gaussian filtering
- Median filtering
- Bilateral filtering
- Morphological operations

Median filtering is particularly useful for removing impulse noise while preserving important edges.

2. Handling Partial Occlusions

When a portion of the component is hidden, the system should not immediately classify the component as defective.

Possible approaches include:

- PCA shape priors
- Partial contour matching
- Feature-based matching
- Robust feature extraction
- Image completion or segmentation methods

- Training models with partially occluded examples

PCA-based shape models are useful because the system can estimate the expected complete shape from the visible portion.

3. Handling Varying Illumination

Changes in lighting can significantly affect image intensity. The following techniques can be used:

- Histogram equalization
- CLAHE
- Intensity normalization
- Adaptive thresholding
- Background correction
- Controlled industrial lighting

4. Data Augmentation

The training dataset can be augmented with variations in:

- Brightness
- Contrast
- Rotation
- Scale
- Noise
- Partial occlusion

This helps the model become more robust to real manufacturing conditions.

Conclusion

A combination of preprocessing, robust shape representation, data augmentation, and controlled image acquisition can significantly improve defect detection under noisy, partially occluded, and unevenly illuminated conditions.

Overall Solution for Manufacturing Defect Detection

Proposed Architecture

Industrial Camera

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Image Acquisition

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Preprocessing & Noise Removal

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Component Segmentation

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Contour Extraction + PCA Shape Model

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Feature Extraction / CNN

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SVM / GMM / CNN Classifier

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Defect Detection

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Accept / Reject Decision

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Quality Control Database

Performance Evaluation

The system can be evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- False Positive Rate
- False Negative Rate
- Processing time per component

For manufacturing, **false negatives are particularly important**, because a defective component incorrectly classified as normal may reach the customer.

2. Autonomous Vehicle Obstacle Recognition

Industry Problem

A self-driving vehicle must detect obstacles such as pedestrians, animals, traffic cones, fallen objects, and other unexpected objects on urban roads. Rare obstacles are difficult to detect because only a small number of labeled examples may be available. The system must therefore use data augmentation and efficient deep-learning models while meeting real-time inference requirements on embedded hardware.

Question 1: GAN-Based Data Augmentation

Answer

Generative Adversarial Networks (GANs) can generate synthetic images that resemble real-world obstacle images.

A GAN consists mainly of two components:

Generator → Creates synthetic images

Discriminator → Determines whether images are real or synthetic

During training, the generator attempts to produce realistic obstacle images while the discriminator learns to distinguish real images from generated images.

Proposed Process

Rare Real Obstacle Images

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GAN Training

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Synthetic Obstacle Generation

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Quality Filtering

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Combine Real + Synthetic Images

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Train Detection Model

For example, if the training dataset contains very few images of a rare obstacle, a GAN can generate additional variations involving different:

- Viewing angles
- Backgrounds
- Object positions
- Lighting conditions
- Scales
- Environmental conditions

These synthetic images can increase dataset diversity and reduce overfitting.

Important Consideration

Synthetic images should not automatically be added to the training dataset. Generated images should be checked for visual quality and correct object characteristics. Poor-quality synthetic images can introduce incorrect patterns and reduce model performance.

Conclusion

GAN-based augmentation is useful for addressing class imbalance and limited training data. It can provide additional examples of rare obstacles and improve the robustness of the detection model when synthetic samples are carefully validated.

Question 2: CNN vs Vision Transformer

Answer

CNNs and Vision Transformers (ViTs) both have advantages for obstacle detection.

Feature	CNN	Vision Transformer
Feature extraction	Strong local feature extraction	Uses attention to model global relationships
Small objects	Good with multi-scale feature pyramids	Can capture wider contextual relationships
Computational cost	Generally efficient	Can require greater computational resources
Training data	Can perform well with moderate datasets	Often benefits from large-scale pretraining
Real-time deployment	More suitable for embedded systems	More demanding depending on architecture
Global context	Limited compared with attention mechanisms	Strong global context modeling
Embedded deployment	Generally easier	Requires careful optimization

CNN Approach

CNN-based object detectors can extract local features such as edges, textures, shapes, and object patterns. Multi-scale feature extraction is particularly useful for detecting small obstacles.

Vision Transformer Approach

Vision Transformers divide an image into patches and use attention mechanisms to learn relationships between different regions. This can help recognize small or sparse objects by considering the surrounding scene.

Comparison

For high-resolution autonomous-driving images, Vision Transformers can provide strong contextual understanding, particularly when long-range relationships are important. However, their computational requirements can be higher.

CNN-based detectors are generally easier to optimize for real-time embedded deployment.

Conclusion

For a resource-constrained autonomous vehicle, an optimized **CNN-based detector with multi-scale feature extraction** is a practical choice. A lightweight Transformer or hybrid CNN-Transformer model can be considered when the embedded hardware has sufficient computational capability.

Question 3: Real-Time Inference Pipeline

Answer

The autonomous vehicle requires a real-time pipeline that balances **accuracy, latency, memory usage, and power consumption**.

Proposed Pipeline

Camera Sensor



Image Capture



Preprocessing



Image Resizing / Normalization



Lightweight Object Detection Model



Multi-Scale Feature Extraction



Obstacle Classification & Localization



Confidence Filtering



Object Tracking



Risk Assessment



Vehicle Decision System

Step 1: Camera Input

The vehicle camera continuously captures frames from the road environment.

Step 2: Preprocessing

Images are resized and normalized to reduce computational requirements while preserving important obstacle information.

Step 3: Lightweight Detection Model

A compact CNN-based detector can be used for real-time detection. The model should be optimized for embedded hardware.

Step 4: Multi-Scale Detection

Small obstacles can be difficult to detect because they occupy only a few pixels. Multi-scale feature extraction helps detect objects at different sizes.

Step 5: Confidence Filtering

Predictions below a selected confidence threshold can be filtered to reduce false detections.

Step 6: Object Tracking

Detected obstacles can be tracked across consecutive frames. Tracking reduces the need to completely rediscover an object in every frame and provides information about its movement.

Step 7: Risk Assessment

The system can estimate:

- Distance from vehicle
- Relative movement
- Position in driving path
- Estimated collision risk

The obstacle information is then passed to the vehicle's decision-making system.

Computational Constraints in Embedded Systems

Since autonomous vehicles require rapid response, the model should be optimized using:

1. Model Quantization

Convert model parameters from higher-precision formats to lower-precision formats such as INT8 where supported, reducing memory and computation.

2. Pruning

Remove less important model parameters to reduce model size and computational requirements.

3. Knowledge Distillation

Train a smaller student model using knowledge from a larger teacher model.

4. Image Resolution Optimization

Use an appropriate input resolution rather than unnecessarily processing every frame at maximum resolution.

5. Hardware Acceleration

Use an embedded GPU, NPU, or other supported accelerator when available.

6. Efficient Frame Processing

Process frames efficiently and use tracking between detections when appropriate.

7. Latency Monitoring

Measure the complete processing time from camera input to detection output.

Analysis and Performance Evaluation

The autonomous obstacle detection system should be evaluated using:

- **Precision**
- **Recall**
- **F1-score**
- **mAP**
- **Detection latency**
- **Frames per second (FPS)**
- **False positive rate**
- **False negative rate**
- **Memory usage**

For autonomous driving, **recall for rare and safety-critical obstacles is particularly important**, because missing an obstacle can have serious consequences.

A practical system should therefore balance detection accuracy with real-time performance rather than maximizing accuracy alone.

Overall Proposed Autonomous Vehicle Architecture

Camera Sensors

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Image Preprocessing

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GAN-Augmented Training Dataset

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CNN / Vision Transformer Detector

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Multi-Scale Obstacle Detection

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Confidence Filtering

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Object Tracking

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Distance & Risk Estimation

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Decision-Making System

Final Comparative Conclusion

Industry Problem	Recommended Approach	Main Benefit
Manufacturing Defect Detection	PCA + Contours + SVM/CNN	Robust detection of defects despite manufacturing variations
Rare Obstacle Detection	GAN + CNN/Transformer	Improves learning from limited rare-object data
Embedded Real-Time Detection	Lightweight CNN + optimization	Provides faster inference with lower computational requirements

Overall Conclusion:

Both industry problems require a combination of computer vision, machine learning, and deep learning techniques. For manufacturing, PCA-based shape priors and contour representations provide a strong method for modeling acceptable component variations, while SVM, GMM, or CNN models can classify defects. For autonomous vehicles, GAN-based augmentation can address limited rare-object data, while CNNs and Vision Transformers provide powerful object-detection capabilities. For real-time deployment, model optimization techniques such as quantization, pruning, knowledge distillation, and hardware acceleration are essential for achieving an effective balance between accuracy and computational efficiency.